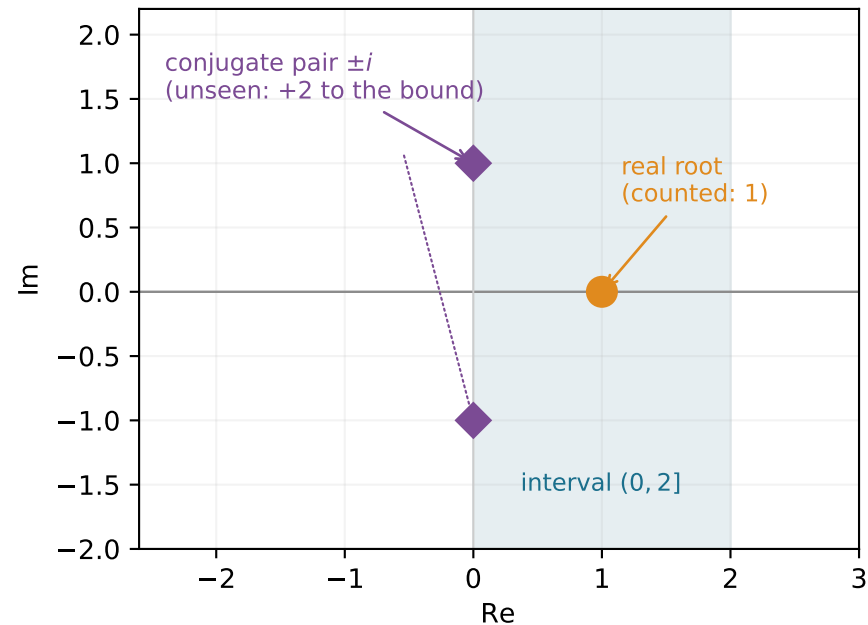
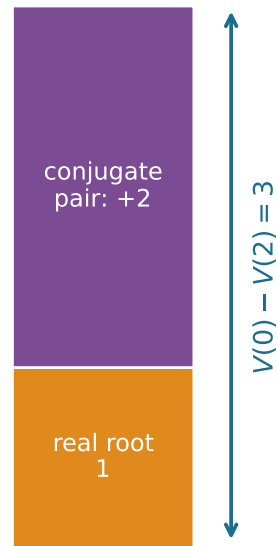


Descartes' shadow: the even surplus is twice the unseen conjugate pairs

Roots of $p = (x^2 + 1)(x - 1)$ in $\mathbb{C}$



the Budan--Fourier bound



Descartes' rule is the shadow on $(0, \infty)$:
coeff. sign changes of $+, -, +, - = 3$,
overcounting by the even number 2.